Lesson 4 · Data in class: record, chart, decide · for your pupils

Our class rainfall chart

40 minutes | Ages 10–16 · adapt for other ages below | ✓ No electricity needed | ✓ No devices needed | ✓ Photocopy-friendly

Big question

How can rain data help us decide when to water?

Pupils will be able to...

- Record a week of rain readings in a table with units.
- Draw a bar chart with equal steps and labelled axes.
- Spot a pattern and use a class rule to make a decision.

You need

- Board and chalk, with a ruler for straight bars
- A week of rain readings: your class gauge log or the sample (see prep)
- Squared paper or exercise books for pairs
- Worksheet: Rain data detective (one per pair, optional)

Before class

Draw the empty table and chart axes on the board. Sample data if you have no gauge yet (8 am readings, made up for practice): Mon 14 mm, Tue 9, Wed 0, Thu 0, Fri 0, Sat 0, Sun 2.

Time	What you do	What you can say	Pupils
Warm-up 0–5 min	Ask: "When should we water the school garden?" Collect answers — every day? When it looks dry? Write "How could data help us decide?" on the board.	*"Today we'll let the rain readings help us decide, not just our guesses."*	Suggest when to water and give reasons.
Record 5–12 min	Pupils copy the table — Day, Time, Rain (mm) — and fill it from the class gauge log or the sample data. Discuss what to write for a day with no reading.	*"Write 0 for a dry day, and "missed" if nobody read the gauge. Why not leave it blank?"*	Copy the table and explain the difference between 0 and "missed".
Chart 12–25 min	Draw the axes together: days along the bottom, "Rain (mm)" up the side in equal steps of 2. Pairs draw their bars on paper while volunteers draw them on the board.	*"Every step on the side must be the same size. Count up carefully before you draw each bar."*	Draw a bar chart in pairs; check each other's bar heights.
Decide 25–35 min	Ask pairs to name one pattern. Then agree a class rule, such as "water if less than 5 mm fell in three days and the soil feels dry". Apply it to the chart.	*"What does the chart tell us? Does our rule say water today, or wait?"*	Name a pattern, vote on the rule and apply it to the chart.
Wrap-up 35–40 min	Explain that connected devices can do these steps automatically with sensors and rules. Exit ticket: the four steps in order, and today's decision.	*"A machine can do this too — but someone still has to check that the data makes sense."*	Write the exit ticket.

Check understanding
Listen for patterns described with data — “four dry days in a row” — not just “it was dry”. Check bars against the table, and that the side has equal steps and “mm”.

Exit ticket
The four data steps in order, and our decision today.

Common mix-ups

Pupils may think…	Help them see…
A blank box means no rain.	A blank could mean no rain or no reading. Write 0 for dry days and “missed” when nobody read the gauge.
The side of the chart can use any numbers.	Steps must be equal — 0, 2, 4, 6. Unequal steps make some bars look bigger or smaller than they are.
The chart tells us exactly what to do.	The chart shows what happened. The class rule — agreed in advance — turns the pattern into a decision.

Adapt it

Younger pupils
Ages 6–9: draw a raindrop on the board for each rainy day and a sun for each dry day. Count them together and ask: water today, or wait?

Older pupils and university
Pupils add a temperature column, draw a second chart and look for a link: are dry days warmer? Discuss why a link doesn’t prove one causes the other.

Large classes
Give each row one day of data. A volunteer from each row draws that day’s bar on the board while the class checks the height.

No board or no space
Make the chart on the ground: each day is a line of stones, one stone per millimetre. Pupils walk along it to spot the pattern.

Short on time? Put the sample data on the board, draw the chart together, and agree the watering decision as a class (20 minutes).

What you need to know

Readings only help when you use them: record them in a table, draw a chart, spot the pattern, then decide. Connected devices follow the same four steps automatically.

Data
Facts collected to answer a question, such as a week of rainfall readings. Each reading is one piece of data.

Table
Rows and columns that keep readings in order. Each column has a heading with its unit, such as “Rain (mm)”.

Bar chart
A picture of data where each bar’s height shows a number. Tall and short bars make patterns easy to see.

Pattern
A trend or repeat you can see in data, such as rain on most afternoons or a run of dry days.

My notes — how did it go?